

circ_bfr.vr

File location: \$VCS_HOME/doc/examples/nativetestbench/openvera/rvm/coverage/circ_bfr/circ_bfr.vr

```
class circular_buffer {
  protected bit [3:0] rd_ptr;
  protected bit [3:0] wr_ptr;
  local integer bfr[16];
  task new() {
    this.reset();
  }
  task reset() {
    this.rd_ptr = 0;
    this.wr_ptr = 0;
  }
  task write(integer v) {
    if (!this.is_full()) this.bfr[this.wr_ptr++] = v;
  }
  function integer read() {
    read = 'bX;
    if (!this.is_empty()) read = this.bfr[this.rd_ptr++];
  }
  local function integer is_full() {
    is_full = (wr_ptr+1 == rd_ptr);
  }
  local function integer is_empty() {
    is_empty = (wr_ptr == rd_ptr);
  }
}
```